Rotating Workforce Scheduling

Problem model, counting conventions and report scope

STATE

A partial 28-day cyclic sequence X[0]...X[27]. Each position has domain D (day), A (afternoon), N (night), O (off); ? means unassigned.

INITIAL STATE

All 28 positions unassigned. Four employees use weekly offsets of the same four-week sequence.

ACTIONS

Assign the next chronological position, trying D, A, N, O in that fixed order.

SEARCH RESULT

DFS calls (valid states): 498
Tried assignments: 1948
Rejected immediately: 1451
Backtracked assignments: 1920
Solution decisions: 28

CONSTRAINTS

Mon-Sat: one employee each on D/A/N/O. Sunday: D=1, A=0, N=1, O=2. Work blocks last 2-6 days, rest blocks 1-3; N followed by D is forbidden. Include the cyclic boundary.

GOAL

A complete sequence whose four employee rotations meet all constraints.

STRATEGY

Chronological DFS with backtracking and necessary constraint checks; no MRV or heuristic score. This is the confirmed educational model, not a universal workforce specification.

PAGES THAT FOLLOW

The following pages cover all 28 choices on the returned solution branch. A cumulative tree retains all earlier choices and attempted alternatives, with the current expansion enlarged below. Unsuccessful side subtrees are collapsed and counted; this is not an unabridged 498-call trace. The full attempt trace is saved alongside this report as workforce-report-trace.json.

COUNTING NOTE

Statistics apply only to this instance, action order and goal-test timing. Attempted-transition nodes are not the same as unique discovered states. Green nodes identify the returned route retrospectively.

Decision 1 of 28: assign X[0]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

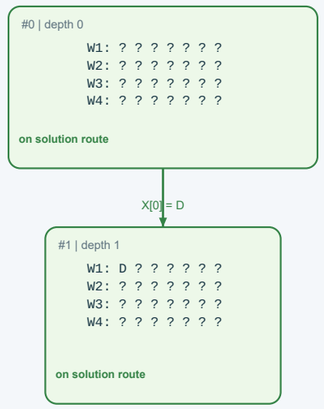
Current depth: 0 | Node #0 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[0] = D (node #1). 0 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

Decision 2 of 28: assign X[1]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

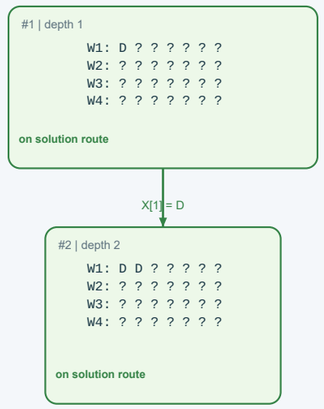
Current depth: 1 | Node #1 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[1] = D (node #2). 0 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

Decision 3 of 28: assign X[2]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

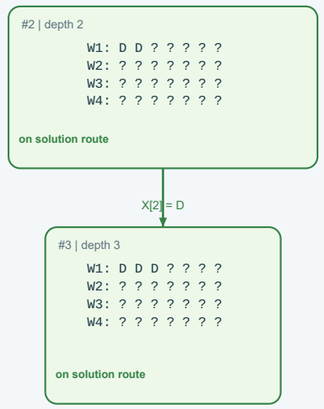
Current depth: 2 | Node #2 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[2] = D (node #3). 0 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

Decision 4 of 28: assign X[3]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

Current depth: 3 | Node #3 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[3] = D (node #4). 0 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

Decision 5 of 28: assign X[4]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

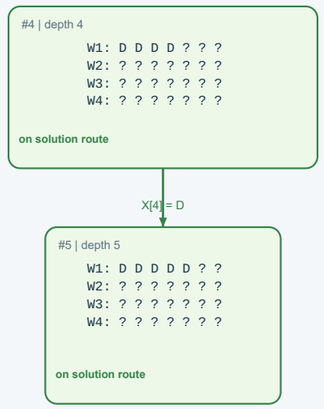
Current depth: 4 | Node #4 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[4] = D (node #5). 0 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

Decision 6 of 28: assign X[5]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

Current depth: 5 | Node #5 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[5] = D (node #6). 0 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

Decision 7 of 28: assign X[6]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

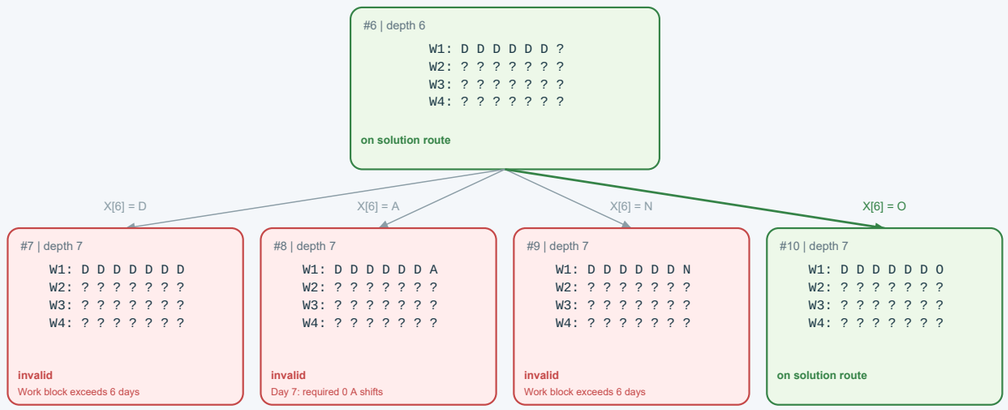
Current depth: 6 | Node #6 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[6] = O (node #10). 3 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

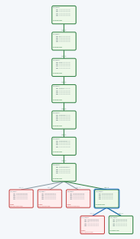
Decision 8 of 28: assign X[7]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

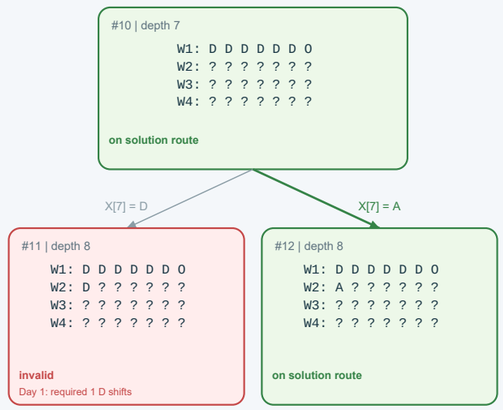
Current depth: 7 | Node #10 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[7] = A (node #12). 1 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

Decision 9 of 28: assign X[8]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

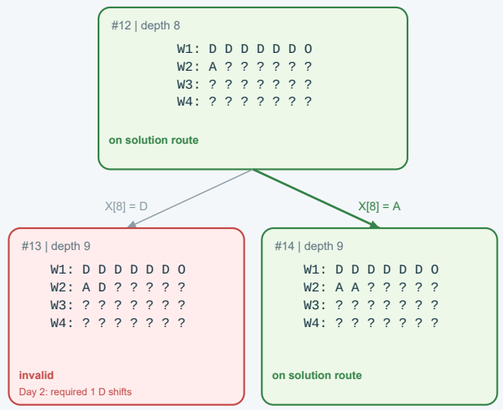
Current depth: 8 | Node #12 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[8] = A (node #14). 1 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

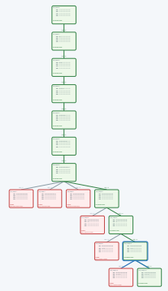
Decision 10 of 28: assign X[9]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

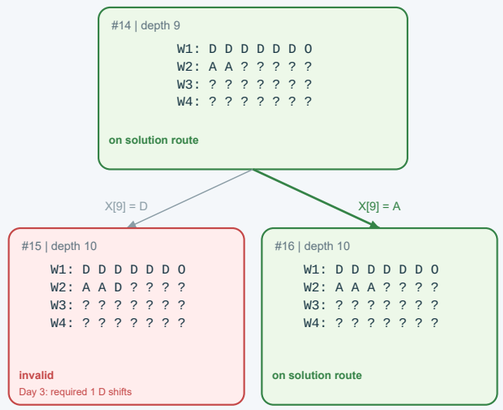
Current depth: 9 | Node #14 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[9] = A (node #16). 1 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

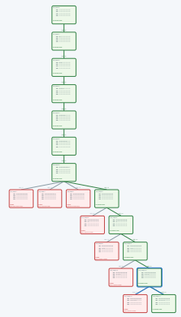
Decision 11 of 28: assign X[10]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

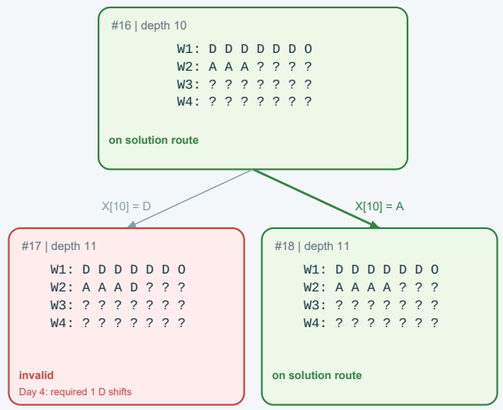
Current depth: 10 | Node #16 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[10] = A (node #18). 1 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

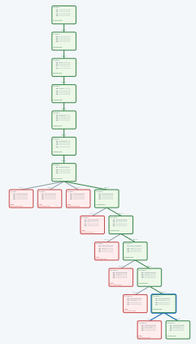
Decision 12 of 28: assign X[11]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

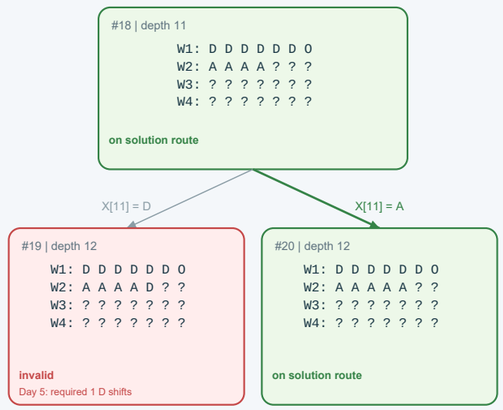
Current depth: 11 | Node #18 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[11] = A (node #20). 1 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

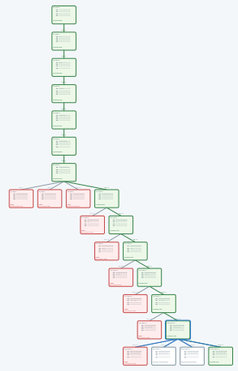
Decision 13 of 28: assign X[12]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

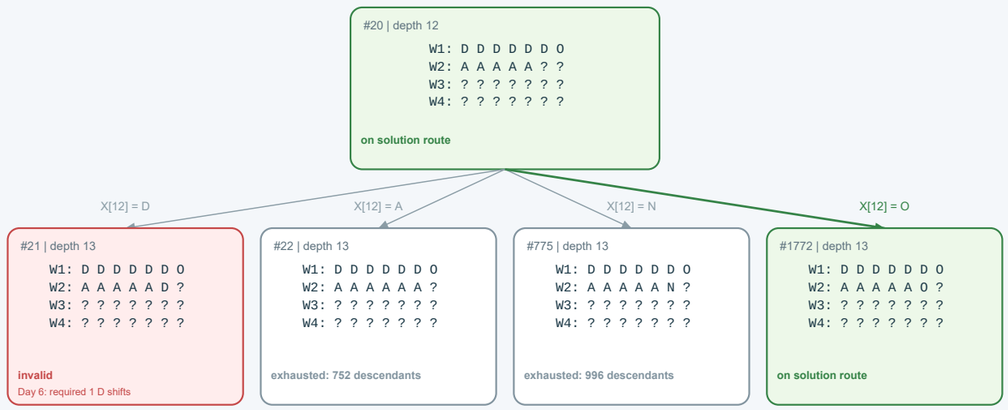
Current depth: 12 | Node #20 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[12] = O (node #1772). 3 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

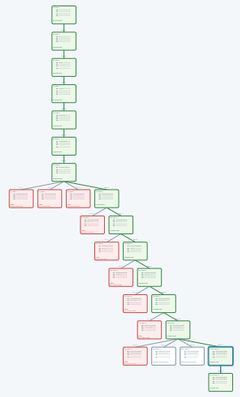
Decision 14 of 28: assign X[13]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

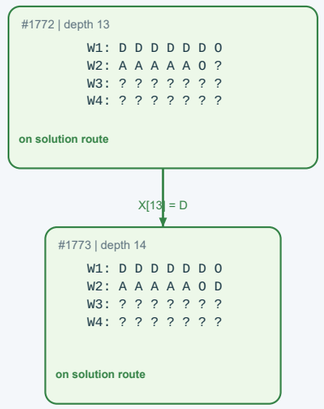
Current depth: 13 | Node #1772 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[13] = D (node #1773). 0 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

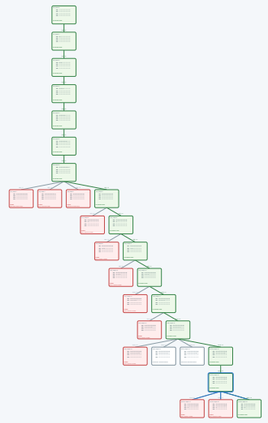
Decision 15 of 28: assign X[14]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

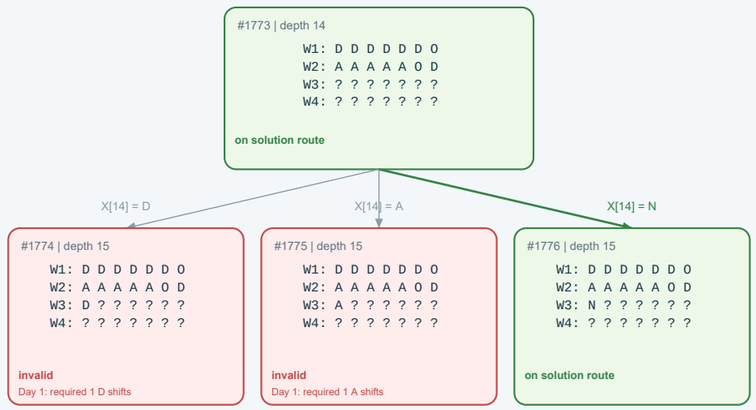
Current depth: 14 | Node #1773 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[14] = N (node #1776). 2 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

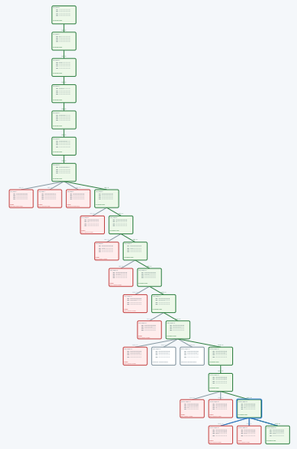
Decision 16 of 28: assign X[15]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

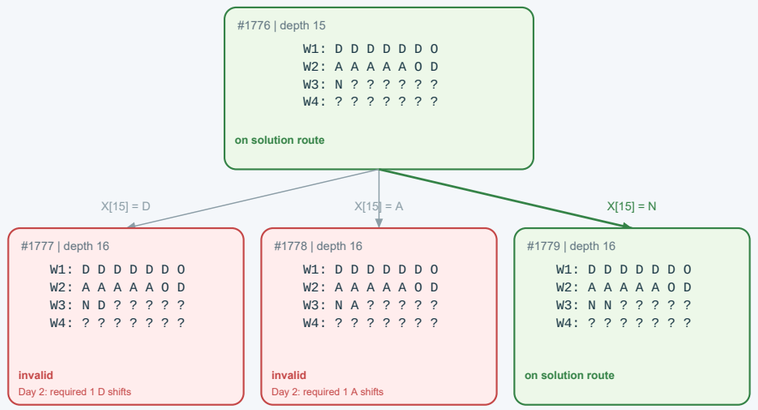
Current depth: 15 | Node #1776 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[15] = N (node #1779). 2 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

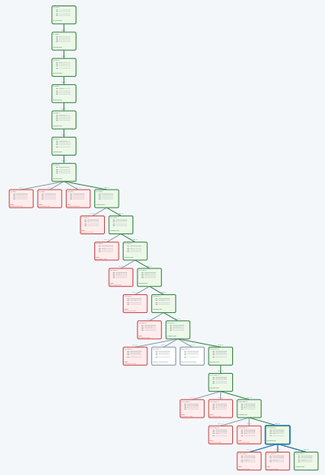
Decision 17 of 28: assign X[16]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

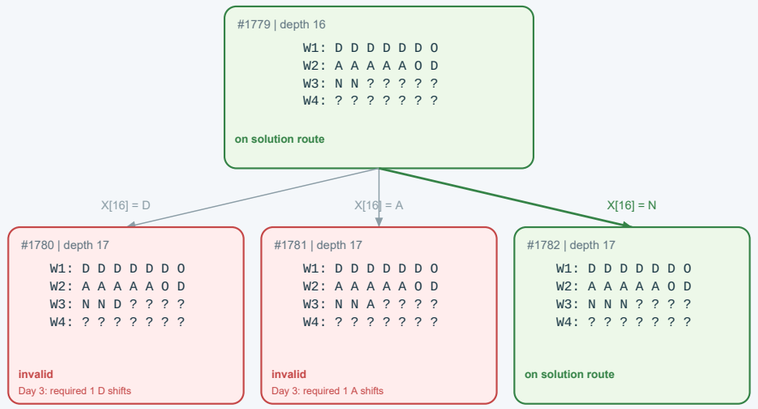
Current depth: 16 | Node #1779 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[16] = N (node #1782). 2 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

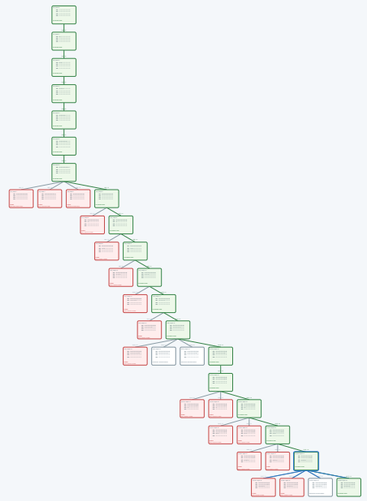
Decision 18 of 28: assign X[17]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

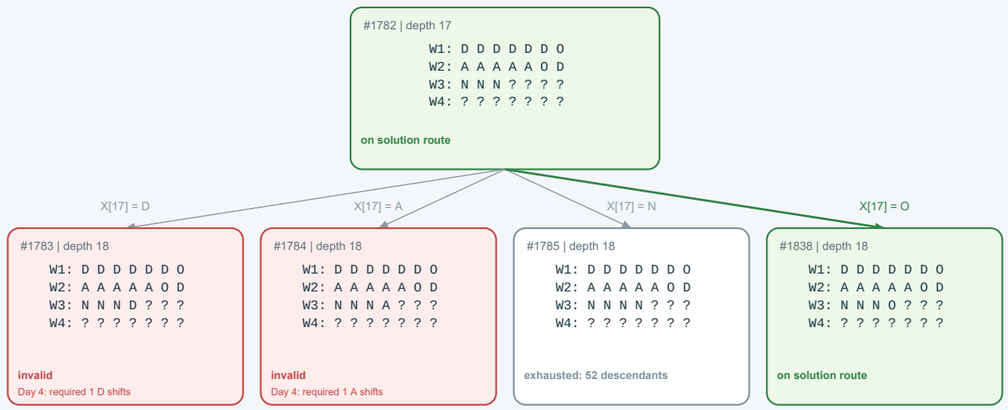
Current depth: 17 | Node #1782 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[17] = O (node #1838). 3 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

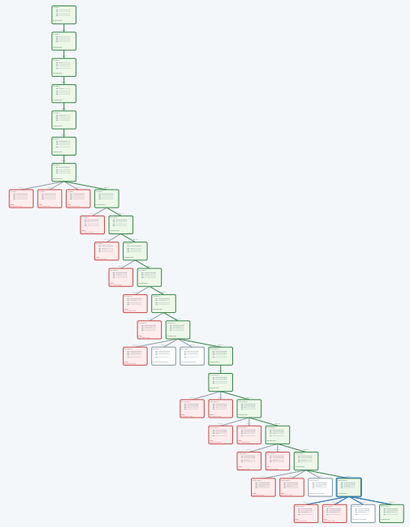
Decision 19 of 28: assign X[18]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

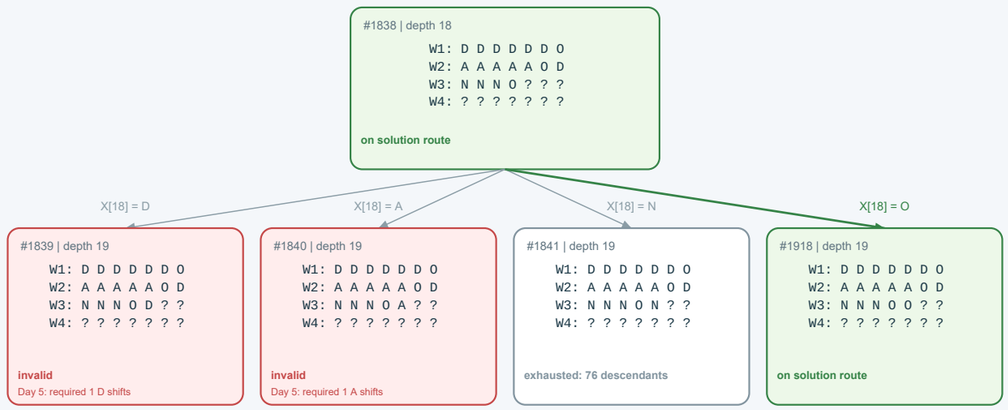
Current depth: 18 | Node #1838 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[18] = O (node #1918). 3 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

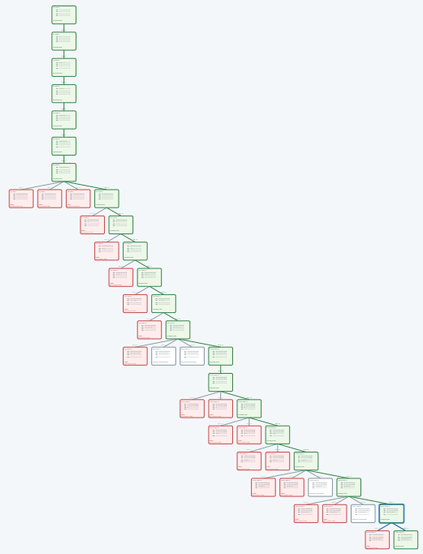
Decision 20 of 28: assign X[19]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

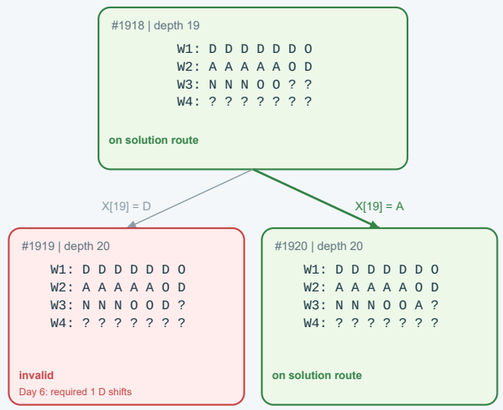
Current depth: 19 | Node #1918 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[19] = A (node #1920). 1 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

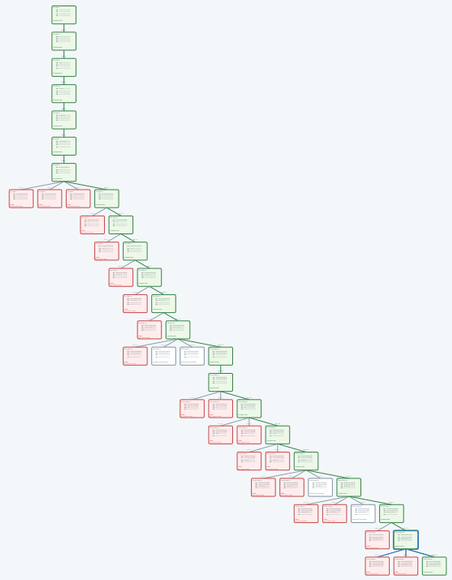
Decision 21 of 28: assign X[20]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

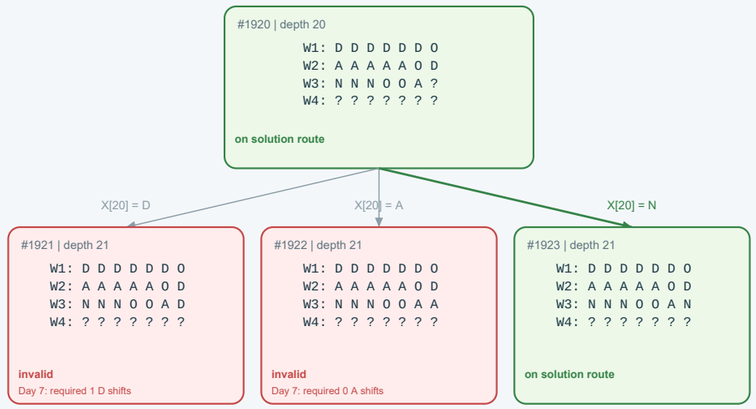
Current depth: 20 | Node #1920 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[20] = N (node #1923). 2 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

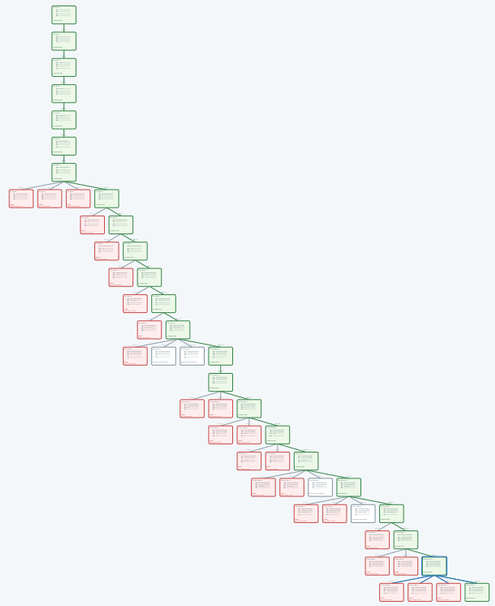
Decision 22 of 28: assign X[21]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

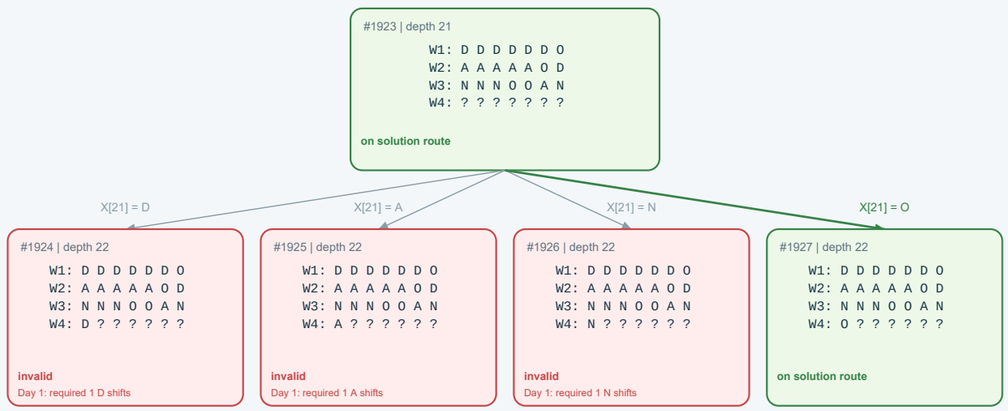
Current depth: 21 | Node #1923 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[21] = O (node #1927). 3 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

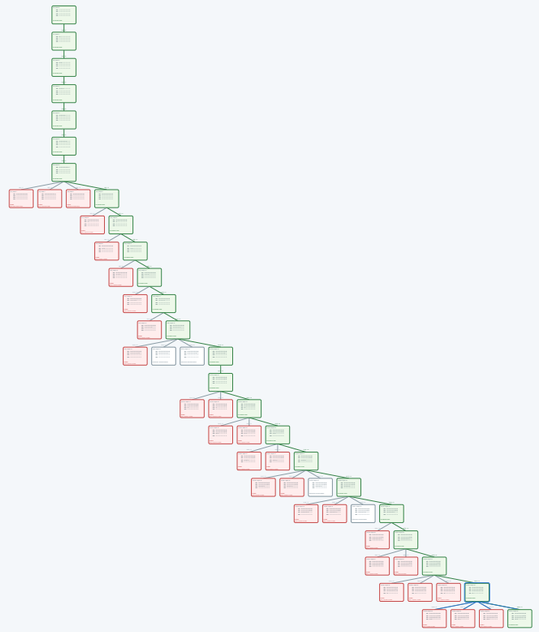
Decision 23 of 28: assign X[22]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

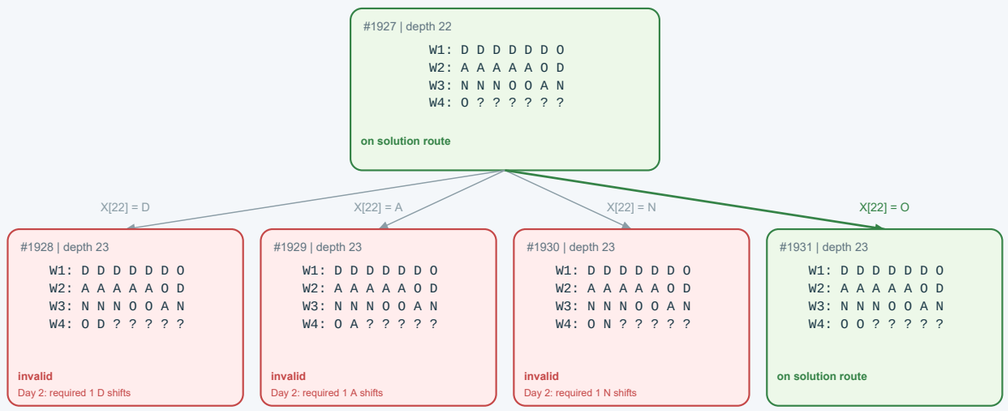
Current depth: 22 | Node #1927 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[22] = O (node #1931). 3 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

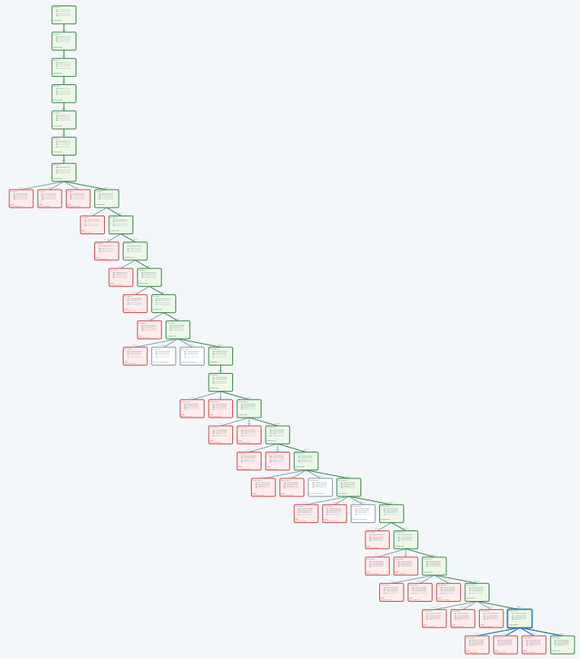
Decision 24 of 28: assign X[23]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

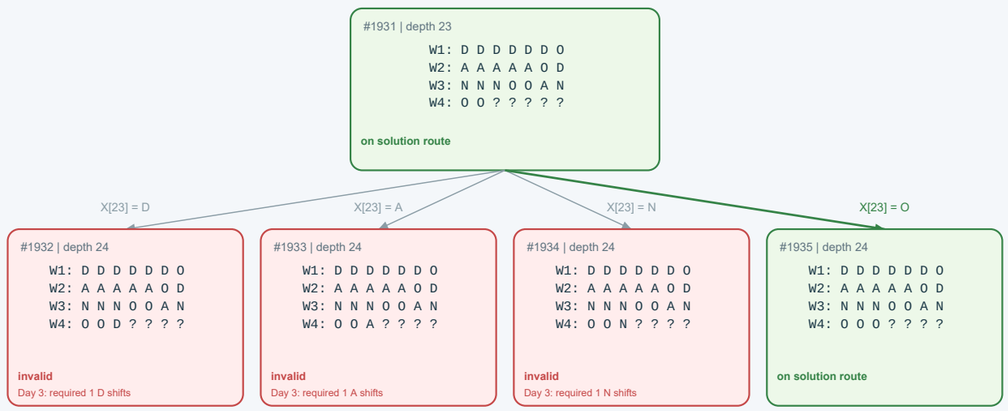
Current depth: 23 | Node #1931 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[23] = O (node #1935). 3 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

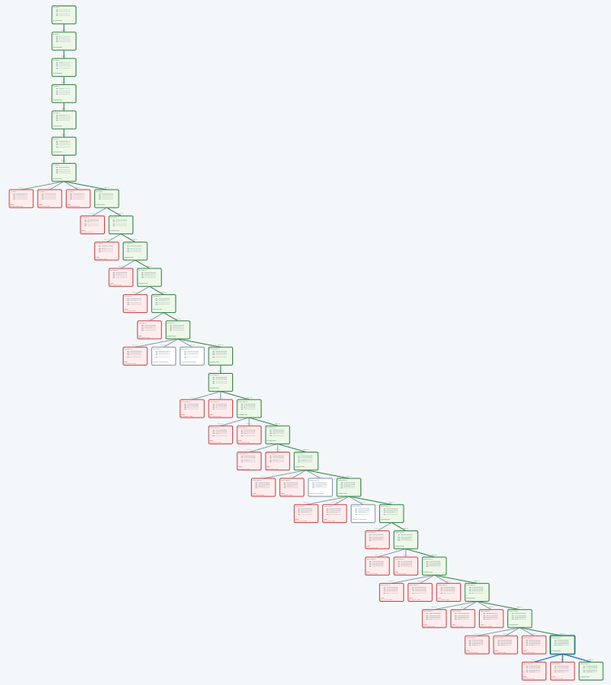
Decision 25 of 28: assign X[24]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

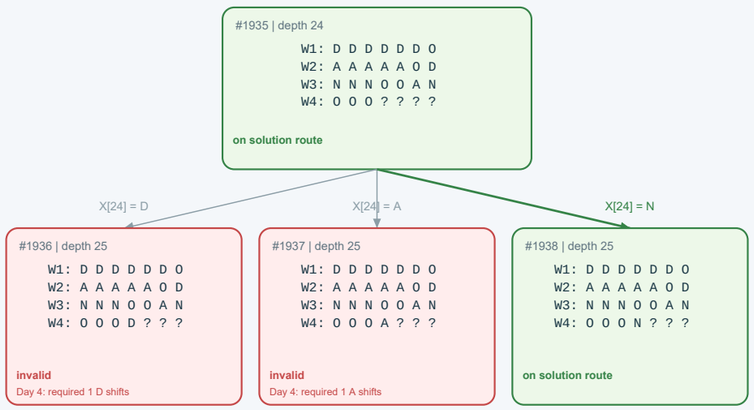
Current depth: 24 | Node #1935 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[24] = N (node #1938). 2 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

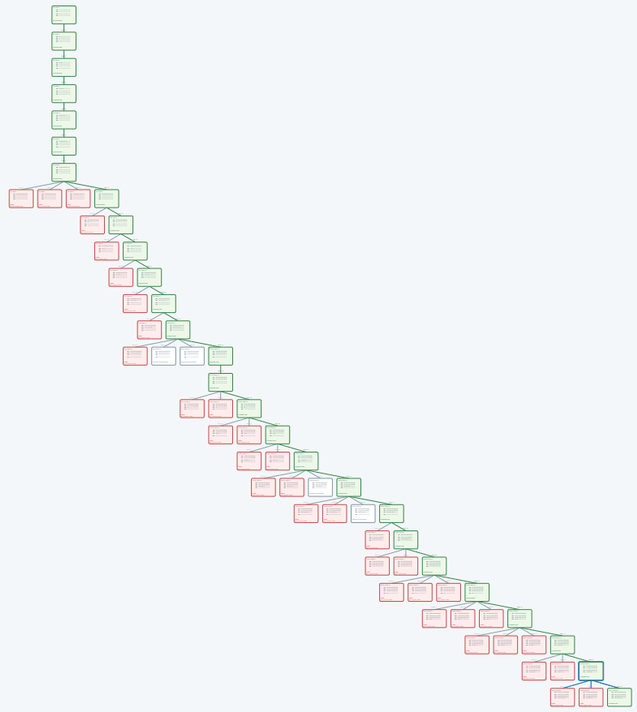
Decision 26 of 28: assign X[25]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

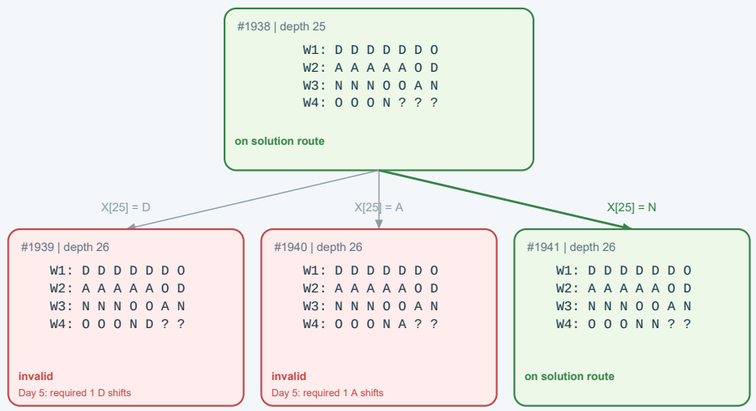
Current depth: 25 | Node #1938 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[25] = N (node #1941). 2 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

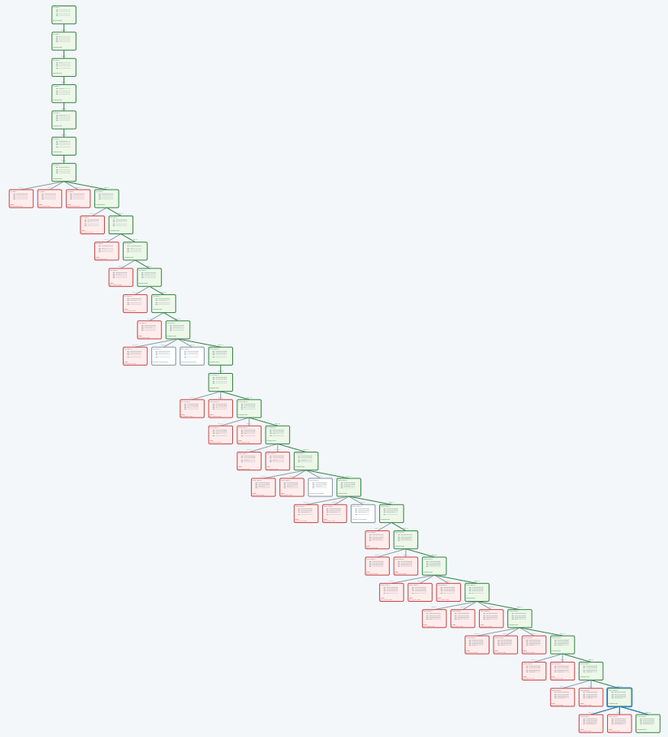
Decision 27 of 28: assign X[26]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

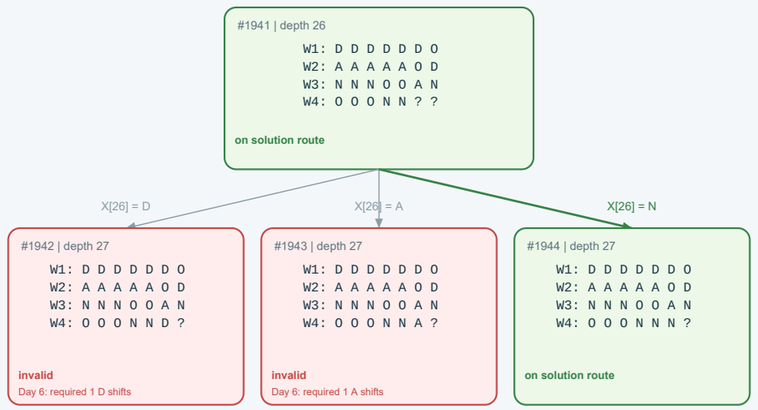
Current depth: 26 | Node #1941 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[26] = N (node #1944). 2 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

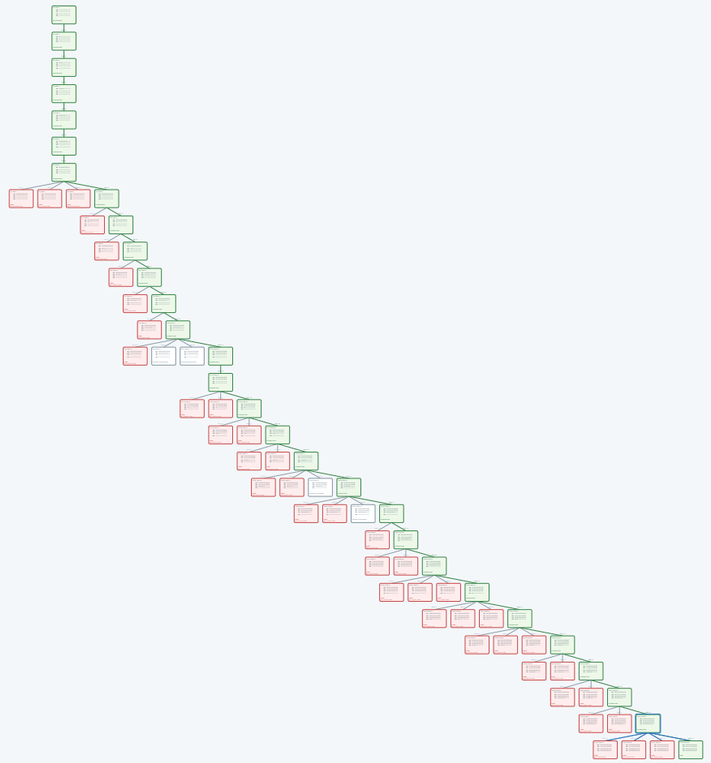
Decision 28 of 28: assign X[27]

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

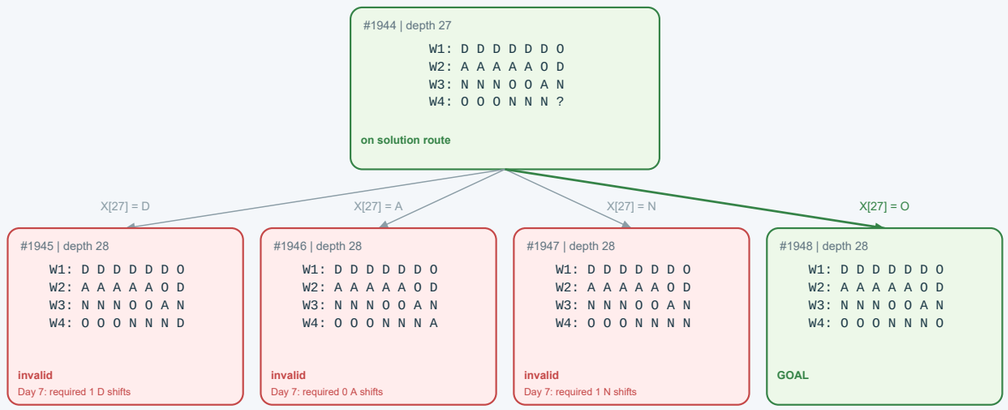
Current depth: 27 | Node #1944 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The returned solution continues with X[27] = O (node #1948). 3 other attempted alternative(s) are shown. Gray exhausted subtrees were searched fully before this choice; their internal nodes are omitted from the report. Values never tried after success are not drawn.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

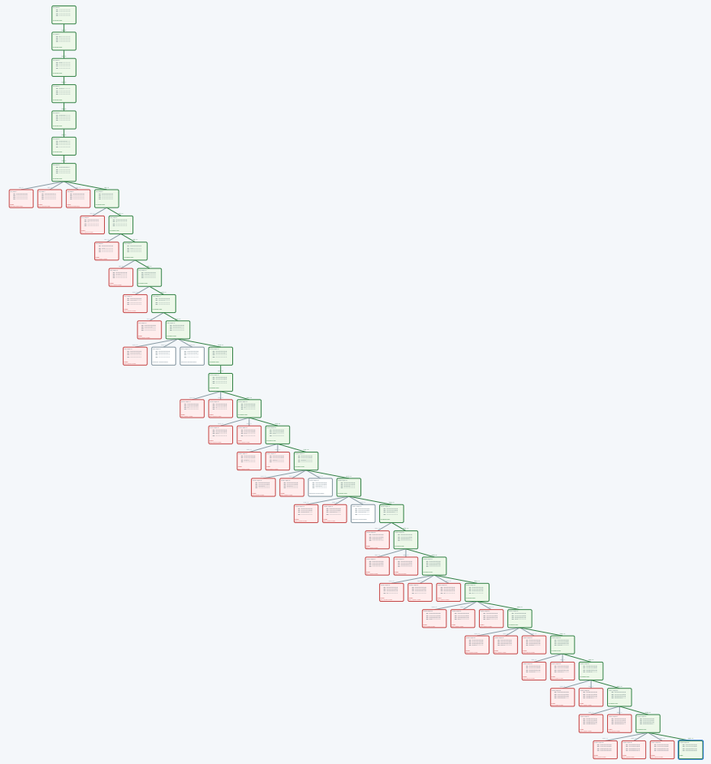
Goal check: all 28 positions assigned

Verified solution branch with actual attempted alternatives; unsuccessful side subtrees are collapsed.

Current depth: 28 | Node #1948 | Fixed value order: D, A, N, O | No heuristic ranking

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above

#1948 | depth 28

```
W1: D D D D D D O
W2: A A A A A O D
W3: N N N O O A N
W4: O O O N N N O
```

GOAL

The complete 28-day sequence satisfies the goal. Recursive calls return success; no further alternatives are tried.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

Final result and verification

The returned solution has been independently checked.

Complete shared cycle

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Sequence week 1	D	D	D	D	D	D	O
Sequence week 2	A	A	A	A	A	O	D
Sequence week 3	N	N	N	O	O	A	N
Sequence week 4	O	O	O	N	N	N	O

Validation: PASS. Coverage was checked for every employee rotation and weekday. Cyclic work/rest run lengths and the N-to-D transition were checked separately. Employee e in week w uses sequence week (e+w) mod 4. The rows above are consecutive weeks of one cycle, not four independently chosen schedules.

Independent cross-check: the generated solution and 1,948 attempted assignments match the existing C++ Basic solver. The 28 decision pages summarize its returned branch; they do not prove an optimal objective value, because this model is a feasibility problem.